

Bugs found: 4

1. Array 'arr' is declared with size 2, but the loop writes to indices 0, 1, and 2, causing an out-of-bounds write.
2. Variable 'x' is used in printf without being initialized, resulting in undefined behavior.
3. Right-shifting by a negative value (-1) is undefined behavior in C.
4. Variable 'y' is assigned a value but never used, making it a dead assignment.